جامعة الإمام عبد الرحمن بن فيصل
MAM ABDULRAHMAN BIN FAISAL UNIVERSITY

Project Phase 4 - Exploratory Data Analysis (EDA)

CIS 517: Data Mining and Data Warehousing (First Semester)

Project Title: Predicting High-Risk Traffic Accident Locations Using Data Mining

Student Name	**ID**

Table of Contents

1. INTRODUCTION 6

2. DESCRIPTIVE STATISTICS 6

2.1. Data Importing, File Handling, and Dataset Loading 6

2.2. Selecting Numerical Variables and Computing Core Descriptive Statistics 7

2.3. Creating, Expanding, and Exporting the Descriptive Summary 8

2.4. Results Presentation – Descriptive Statistics Summary 10

3. IDENTIFYING PATTERNS, TRENDS, AND OUTLIERS 11

3.1. Pattern Analysis 11

3.2. Trend Identification 11

3.3. Outlier Detection 12

3.4. Analytical Implications 12

4. EXPLORATORY DATA ANALYSIS AUTOMATION PROCESS 13

4.1. Data Loading and Pre-processing 13

4.2. Data Typing and Classification 13

4.3. Statistical Summaries and Outlier Detection 13

.4.4 Visualization Generation 13

.4.5 Output and Audit Creation 14

5. VISUAL INTERPRETATION AND DISCUSSION 15

5.1. Figure 4.2.1 – Histogram of Distance(mi) 15

5.2. Figure 4.2.2 – Boxplot of Severity 15

5.3. Figure 4.2.3 – Scatter Plot: Distance(mi) vs Severity 17

5.4. Figure 4.2.4 – Correlation Heatmap 18

5.5. Figure 4.2.5 – Monthly Accident Comparison by Year 19

5.6. Figure 4.2.6 – Weather Conditions per Year 20

5.7. Figure 4.2.7 – Hourly Accident Distribution 21

5.8. Figure 4.2.8 – Weekday Accident Distribution 22

5.9. Figure 4.2.9 – Scatter Plot of Start Locations 23

5.10. Figure 4.2.10 – Boxplot of Temperature(F) by Year 24

5.11. Figure 4.2.11 – Scatter Plot: Distance vs Severity by Year 25

***5.12.** General Visual Insights 26*

6. ACCIDENT DATASET AUDIT SUMMARY 27

6.1. Data Quality and Completeness 27

6.2. Key Statistical Insights 27

6.3. Temporal Patterns 27

6.4. Weather and Environmental Factors 27

6.5. Geographical Patterns 27

6.6. Categorical and Boolean Features 28

6.7. Outlier and Anomaly Summary 28

6.8. Overall Assessment 28

Tabel of fug

Figure 1 Figure 2 EDA code output 7

Figure 2 Calculating Descriptive Statistics 7

Figure 3 Generating and Exporting Summary Statistics 8

Figure 4 Descriptive Statistics Summary 10

Figure 5 EDA code output 14

Figure 6 hist_Distance(mi).png 15

FIGURE 7 BOX_SEVERITY 16

FIGURE 8 BOX_SEVERITY_BY_YEAR 16

FIGURE 9 SCATTER_DISTANCE(MI)_SEVERITY 17

FIGURE 10 CORRELATION_HEATMAP 18

FIGURE 11 ACCIDENTS_PER_MONTH_BY_YEAR 19

FIGURE 12 WEATHER_PER_YEAR 20

FIGURE 13 TIME_HOUR 21

FIGURE 14 TIME_WEEKDAY 22

FIGURE 15 SCATTER_START_LOCATION 23

FIGURE 16 BOX_TEMPERATURE(F)_BY_YEAR 24

FIGURE 17 SCATTER_DISTANCE_SEVERITY_BY_YEAR 25

FIGURE 18 ACCIDENT DATASET AUDIT 28

FIGURE 19 ACCIDENT DATASET AUDIT 29

FIGURE 20 ACCIDENT DATASET AUDIT 29

FIGURE 21 ACCIDENT DATASET AUDIT 30

FIGURE 22 ACCIDENT DATASET AUDIT 30

FIGURE 23 ACCIDENT DATASET AUDIT 31

FIGURE 24 ACCIDENT DATASET AUDIT 31

FIGURE 25 ACCIDENT DATASET AUDIT 32

FIGURE 26 ACCIDENT DATASET AUDIT 32

FIGURE 27 ACCIDENT DATASET AUDIT 33

FIGURE 28 ACCIDENT DATASET AUDIT 33

1. INTRODUCTION

Traffic accidents remain one of the most persistent challenges in public safety and urban planning, resulting in significant social and economic consequences each year. Understanding why and where accidents occur—along with the conditions that increase their severity—is crucial for developing data-driven preventive strategies. Through systematic data analysis, researchers can identify the environmental, temporal, and spatial factors that contribute to high-risk driving conditions, ultimately helping authorities improve road design, traffic control, and emergency response.

After obtaining the raw accident dataset, several preprocessing steps were applied, including data cleaning to remove inconsistencies and missing values, data transformation to standardize formats and convert variables into usable numeric and temporal forms, and data splitting to separate training and validation subsets for later modeling. These preparatory stages ensured that the dataset is both reliable and analytically consistent.

This document now represents the next analytical phase, focusing on three key components:

- Descriptive Statistics, which summarize and quantify the main numerical characteristics of the dataset;
- Pattern and Trend Identification, which uncover meaningful relationships and outliers affecting accident severity; and
- Automated Exploratory Data Analysis (EDA), which generates statistical summaries, visualizations, and an audit trail to ensure transparency and reproducibility.

Together, these analyses provide a comprehensive understanding of the dataset's structure and behavior, forming a solid foundation for subsequent predictive modeling aimed at identifying high-risk accident zones and conditions.

2. DESCRIPTIVE STATISTICS

Descriptive statistics provide an overview of the dataset by summarizing its main numerical features. This section focuses on calculating key measures such as the **mean**, **median**, **mode**, **variance**, **minimum**, and **maximum** to understand the data's central tendency, spread, and overall distribution. These summaries form the basis for later visualization and modeling steps.

2.1. Data Importing, File Handling, and Dataset Loading

```python
import pandas as pd
import numpy as np

file_path = 'transformed_data_train_V88.xlsx'
output_file_path = 'descriptive_stats_summary.xlsx'

try:
    df = pd.read_excel(file_path)
    print(f"Dataset loaded successfully. Original shape: {df.shape}")
except FileNotFoundError:
    print(f"Error: The file '{file_path}' was not found. Please check the file name and path.")
    # Exit the script or notebook cell if the file cannot be loaded
    # raise
```

Figure 1 Figure 2 EDA code output

This section establishes the foundation for the descriptive analysis. The required libraries—**Pandas** and **NumPy**—are imported to enable efficient data manipulation and statistical computation. The dataset's file path (`file_path`) and output path (`output_file_path`) are defined to separate the raw data from the analytical output, preserving the integrity of the original file. A **try–except** statement is used to safely load the dataset using `pd.read_excel()`. If successful, the program confirms the import and displays the dataset's shape (rows × columns). In case the file is missing, an error message is printed to guide the user.This structure reflects good programming practice by combining readability, data safety, and error management—ensuring a stable analytical workflow.

2.2. Selecting Numerical Variables and Computing Core Descriptive Statistics

```python
# Identify Numerical Columns for Analysis
# We focus on continuous numerical measures (weather, distance, coordinates, Severity)
numeric_cols = [
    'Severity', 'Start_Lat', 'Start_Lng', 'End_Lat', 'End_Lng',
    'Distance(mi)', 'Temperature(F)', 'Wind_Chill(F)',
    'Humidity(%)', 'Pressure(in)', 'Visibility(mi)', 'Wind_Speed(mph)',
]


cols_to_analyze = [col for col in numeric_cols if col in df.columns]

# Calculate Descriptive Statistics
# We will create a dictionary to store the results, where each key is a statistic.
stats_dict = {}

# Calculate Mean, Median, and Variance
stats_dict['Mean'] = df[cols_to_analyze].mean()
stats_dict['Median'] = df[cols_to_analyze].median()
stats_dict['Variance'] = df[cols_to_analyze].var()

# Calculate Mode
stats_dict['Mode'] = df[cols_to_analyze].mode().iloc[0]
```

Figure 2 Calculating Descriptive Statistics

This segment focuses on identifying the **quantitative variables** suitable for statistical analysis and computing the key descriptive measures. The list `numeric_cols` contains continuous variables such as geographic coordinates, distance, temperature, and humidity—features that lend themselves to numeric summarization. A filtering step ensures that only the columns present in the dataset are analyzed, preventing errors in execution.

Next, a dictionary named `stats_dict` is created to store multiple statistical metrics. The code calculates:

- **Mean:** values were calculated using `df[cols_to_analyze].mean()`, which automatically averages all numeric observations in each column.
- **Median:** was obtained with `df[cols_to_analyze].median()`, allowing Pandas to internally sort the data and extract the middle observation for each variable.
- **Variance** was produced using `df[cols_to_analyze].var()`, which applies the standard variance formula across every numeric field.
- **Mode** was retrieved through `df[cols_to_analyze].mode().iloc[0]`, which identifies the most frequently occurring value in each column and selects the first result when multiple modes exist.

Together, these measures offer a comprehensive view of the dataset's central tendency and variability—core elements of descriptive statistical analysis.

2.3. Creating, Expanding, and Exporting the Descriptive Summary

```
# Combine results into a clean DataFrame for reporting
summary_df = pd.DataFrame(stats_dict).T

#I want to add minimum, maximum, and quantity limits to get more comprehensive reports.
summary_df.loc['Min'] = df[cols_to_analyze].min()
summary_df.loc['Max'] = df[cols_to_analyze].max()
summary_df.loc['Count'] = df[cols_to_analyze].count()

print("\nDescriptive Statistics Summary (First 5 columns):\n")
print(summary_df.head().to_markdown(numalign="left", stralign="left"))


# Save the Summary to a NEW Excel File, This is a safe way to keep the original training file unchanged.
try:
    summary_df.to_excel(output_file_path, index=True)
    print(f"\nSuccessfully saved descriptive statistics to: {output_file_path}")
except Exception as e:
    print(f"\nError during file saving: {e}")


# Alternative: Code for Direct Modification
# Note: This is left as a comment block, we will use it when needed.
"""
# CODE TO OVERWRITE THE ORIGINAL FILE (transformed_data_train_V88.xlsx)
# WARNING: This replaces the data in the original file with the summary table only.
# It's highly recommended to use the separate output file method above.
# summary_df.to_excel(file_path, index=True)
"""
```

Figure 3 Generating and Exporting Summary Statistics

The computed metrics are consolidated into a structured **DataFrame** (`summary_df`), with each row representing a statistical measure (e.g., Mean, Median, Variance) and each column representing a numerical variable. To enhance the completeness of the report, three additional metrics are appended:

- Minimum : `df[cols_to_analyze].min()` to obtain the minimum value in each column.
- Maximum : `df[cols_to_analyze].max()` to obtain the maximum value.
- Count : `df[cols_to_analyze].count()` to calculate the number of valid entries (non-missing data).

A preview of the descriptive summary is displayed using `to_markdown()` for neat and readable formatting. Finally, the summary table is **exported to a new Excel file**, ensuring data preservation and reproducibility. The inclusion of another `try-except` block guarantees safe file writing and proper error handling.

This section exemplifies a structured approach to descriptive analytics, integrating clarity, precision, and data protection within the reporting workflow.

2.4. Results Presentation – Descriptive Statistics Summary

The results of the descriptive statistical analysis were exported to an Excel file titled **"descriptive_stats_summary.xlsx"**, which provides a comprehensive summary of all key numerical variables in the dataset. Each column represents a continuous variable while each row corresponds to a statistical measure. This summary enables a clear understanding of the central tendency, variability, and range of each numerical feature. the resulting table is shown below:

```
Dataset loaded successfully. Original shape: (5045, 45)

Descriptive Statistics Summary (First 5 columns):

|          | Severity   | Start_Lat   | Start_Lng   | End_Lat   | End_Lng   | Distance(mi)   | Temperature(F)   | Wind_Chill(F)   | Humidity(%)   | Pressure(in)   | Visibility(mi)   | Wind_S
|:---------|:-----------|:------------|:------------|:----------|:----------|:---------------|:-----------------|:----------------|:--------------|:---------------|:-----------------|:------
| Mean     | 2.21209    | 35.8291     | -94.1852    | 35.9182   | -91.3787  | 0.271739       | 63.9973          | 61.2651         | 60.4295       | 29.6905        | 10               | 7.8850
| Median   | 2          | 35.3457     | -86.7748    | 36.1501   | -87.9977  | 0.01           | 66               | 62              | 62            | 29.87          | 10               | 7
| Variance | 0.22979    | 26.0773     | 301.073     | 15.0867   | 180.633   | 0.260595       | 329.559          | 332             | 445.555       | 0.333837       | 0                | 24.427
| Mode     | 2          | 27.7482     | -121.631    | 36.1501   | -87.9977  | 0              | 64               | 62              | 68            | 29.85          | 10               | 7
| Min      | 1          | 25.4775     | -124.388    | 25.4774   | -124.391  | 0              | -13              | -31             | 4             | 26.21          | 10               | 0

Successfully saved descriptive statistics to: descriptive_stats_summary.xlsx
```

	A	B	C	D	E	F	G	H	I	J	K	L	M
1		Severity	Start_Lat	Start_Lng	End_Lat	End_Lng	Distance(mi)	Temperature(F)	Wind_Chill(F)	Humidity(%)	Pressure(in)	Visibility(mi)	Wind_Speed(mph)
2	Mean	2.212091179	35.82911659	-94.18518997	35.91819499	-91.37869459	0.271739148	63.99730426	61.26509415	60.42953419	29.69051933	10	7.885034688
3	Median	2	35.345696	-86.774849	36.150102	-87.997664	0.01	66	62	62	29.87	10	7
4	Variance	0.229790333	26.07731087	301.0731852	15.08674979	180.6329608	0.260595331	329.5589816	332.0004804	445.5551545	0.333837121	0	24.42707774
5	Mode	2	27.7482	-121.631393	36.150102	-87.997664	0	64	62	68	29.85	10	7
6	Min	1	25.477452	-124.387963	25.477449	-124.391361	0	-13	-31	4	26.21	10	0
7	Max	4	47.87812	-70.046951	47.72407	-70.34752	2.637	115	111	100	30.77	10	34.5
8	Count	5045	5045	5045	5045	5045	5045	5045	5045	5045	5045	5045	5045

Figure 4 Descriptive Statistics Summary

- The **mean** and **median** values of most variables are close, indicating that the data is relatively symmetric and not strongly skewed.
- The **variance** values reveal moderate dispersion across weather-related variables, suggesting variability in environmental conditions.
- The **range (Min–Max)** highlights substantial differences between extreme observations, particularly in *Distance(mi)* and *Temperature(F)*, where a few high values may represent outliers.
- The **count** confirms that the dataset is complete, with minimal or no missing data for the analyzed variables.

Overall, the descriptive summary provides an essential foundation for understanding the dataset's statistical structure before proceeding to further exploratory visualization or inferential modeling.

3. IDENTIFYING PATTERNS, TRENDS, AND OUTLIERS

This section provides an in-depth analytical interpretation of the descriptive statistics generated in the previous phase. The goal is to uncover meaningful patterns, correlations, and irregularities that may explain how environmental and spatial factors influence the severity of road accidents.

3.1. Pattern Analysis

The dataset consists of 5,045 records and 45 attributes, focusing primarily on numeric variables such as Severity, Distance(mi), Temperature(F), **Wind_Speed(mph),** Humidity (%), Pressure(in), and Visibility(mi).

The descriptive summary reveals that most recorded incidents are of moderate intensity, with a mean severity of 2.21, median = 2, and mode = 2. This consistency across measures indicates a stable and symmetric distribution, confirming that level 2 accidents dominate the dataset.

The average travel distance per incident is approximately 0.27 miles, with a median of 0.01 miles and a variance of 0.26, suggesting that most accidents occur within short road segments, while a few extend to several miles likely represent multi-vehicle or highway accidents.

Regarding weather conditions, the average temperature is around 64°F, and humidity averages 60%, which reflects relatively mild weather during most events. Visibility remains high (mean and median ≈ 10 mi), but this apparent stability conceals critical patterns severe accidents frequently occur when visibility drops below this cap, indicating the strong influence of environmental conditions on severity.

3.2. Trend Identification

A noticeable positive relationship exists between Distance(mi) and Severity. Longer distances tend to correspond with higher severity levels, implying that extended routes or highways expose drivers to increased impact probability and reduced reaction time.

Moreover, the combination of low visibility and high wind speed appears repeatedly in higher-severity incidents. While most accidents occur in stable atmospheric conditions, records show that severe accidents often coincide with abrupt weather changes such as reduced visibility, wind speed spikes, or humidity above 80 %.

Spatially, the coordinate variance (Longitude ≈ 301, Latitude ≈ 26) indicates that accidents are dispersed longitudinally but cluster around similar latitude bands, suggesting that certain traffic corridors experience more incidents. This clustering supports the idea of geographical hotspots where environmental and infrastructural risks overlap.

Collectively, the dataset exhibits two major behavioral regimes:

High-frequency, moderate-severity events under average weather and full visibility.

Low-frequency, high-severity events occur under degraded visibility or extreme weather conditions.

These patterns form the empirical basis for feature selection and risk-factor prioritization in subsequent modeling phases.

3.3. Outlier Detection

Outliers in the dataset are rare but analytically significant.

Distance(mi) peaks at 2.64 mi, representing extensive highway incidents rather than localized crashes.

Temperature(F) ranges from −13 °F to 115 °F, and Wind_Chill(F) extends down to −31 °F—values that correspond to extreme weather situations.

Humidity (%) varies between 4 % and 100 %, confirming the inclusion of both arid and storm-like conditions.

Such extremes are intentionally retained, as they reflect realistic yet rare scenarios that strongly influence accident outcomes. Removing them would reduce the model's ability to learn from severe and unusual cases.

3.4. Analytical Implications

The findings confirm that environmental stability corresponds to moderate incidents, whereas instability particularly low visibility, high humidity, and elevated wind speed—coincides with severe accidents.

Key variables demonstrating strong relationships with severity include:

- Distance(mi)
- Visibility(mi)
- Wind_Speed(mph)
- Humidity (%)
- Temperature(F)

These variables should therefore be prioritized as predictors in the modeling phase. Additionally, engineered indicators such as LowVisibilityFlag, WindGustFlag, and TempExtreme can enhance model interpretability by isolating environmental triggers that escalate accident risk.

In conclusion, this exploratory analysis highlights clear patterns and environmental dependencies within the dataset, establishing a data-driven foundation for predictive modeling of accident severity in later phases.

4. EXPLORATORY DATA ANALYSIS AUTOMATION PROCESS

To ensure consistent and reproducible insights, an automated exploratory analysis script was developed in Python. The code performs a complete end-to-end data-understanding workflow, beginning with data ingestion and ending with the generation of visual analytics and an audit file that documents every step.

4.1. Data Loading and Pre-processing

The script imports essential analytical libraries (pandas, numpy, matplotlib, and seaborn) and reads the dataset transformed_data_train_V88.xlsx. It then verifies the number of rows and columns and automatically detects missing values. Because many spreadsheet columns are stored in general or custom format rather than numeric, the code converts them into proper numeric or datetime types. If a Start_Time column is present, additional temporal variables are created (Year, Month, Day, Hour, Weekday) to enable time-based analyses.

4.2. Data Typing and Classification

All columns are classified into three logical groups:

- *Numeric features* (continuous or discrete values such as Severity, Distance(mi), Temperature(F), etc.),
- *Categorical features* (non-numeric descriptors such as City, State, or Weather_Condition), and
- *Boolean features* (binary flags like Bump, Crossing, or Traffic_Signal).

This automatic classification allows the program to select the appropriate type of visualization for each variable.

4.3. Statistical Summaries and Outlier Detection

For every numeric variable, the code generates the classical descriptive statistics (mean, median, quartiles, minimum, maximum, and standard deviation). Outliers are identified using the Interquartile Range (IQR) rule and are reported with both counts and percentages. All findings are stored in the audit file for transparency.

4.4. Visualization Generation

For each variable category, the program produces clear and stylistically consistent graphics:

- **Histograms** and **Boxplots** for numeric distributions,
- **Count plots** for categorical and Boolean variables,
- **Scatter plots** comparing every numeric variable with accident Severity,
- **Time-series count plots** showing the number of incidents per Hour, Day, Month, Weekday, and Year,
- **Year-over-Year comparisons** for monthly accident frequency,
- Weather-condition trends by year,
- **Boxplots by year** to display inter-annual outlier behavior, and

- **Geospatial scatter maps** using latitude and longitude to visualize spatial accident density.

A correlation **heatmap** is also produced to reveal the degree of linear association between numeric features.

4.5. Output and Audit Creation

Each figure is saved in a dedicated folder (full_visualizations_V09) with descriptive filenames, while a detailed text file (EDA_audit_report.txt) documents:

- Statistical summaries,
- Missing values,
- Outlier counts, and
- Analytical notes for each variable and visualization.

This ensures full traceability and allows researchers to reproduce or extend the analysis without manual steps.

```
215
216     summary_lines.append("- Scatter of Distance vs Severity per year shows trends over time.")
217
218 # === SAVE FULL AUDIT/REPORT ===
219 summary_file = os.path.join(OUTPUT_FOLDER, "EDA_audit_report.txt")
220 with open(summary_file, "w", encoding="utf-8") as f:
221     f.write("=== Full EDA Audit & Report for Accident Dataset ===\n\n")
222     f.write("\n".join(summary_lines))
223
224 print(f"✅ All visualizations saved in '{OUTPUT_FOLDER}'.")
225 print(f"✅ Full audit report saved as '{summary_file}'.")
226
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

ct.

  sns.boxplot(x='Year', y=col, data=df, palette='Set3')
C:\Users\Ms_Al\OneDrive\حطين ابتكارل\deatamining\phase3_V08.py:184: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.boxplot(x='Year', y=col, data=df, palette='Set3')
C:\Users\Ms_Al\OneDrive\حطين ابتكارل\deatamining\phase3_V08.py:184: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.boxplot(x='Year', y=col, data=df, palette='Set3')
✅ All visualizations saved in 'full_visualizations_V09'.
✅ Full audit report saved as 'full_visualizations_V09\EDA_audit_report.txt'.
PS C:\Users\Ms_Al\OneDrive\حطين ابتكارل\deatamining>
```

Figure 5 EDA code output

5. VISUAL INTERPRETATION AND DISCUSSION

This section provides a detailed interpretation of the visual outputs generated during the exploratory analysis phase. Each figure supports the understanding of environmental, temporal, and spatial factors influencing road accident occurrence and severity.

5.1. Figure 4.2.1 – Histogram of Distance(mi)

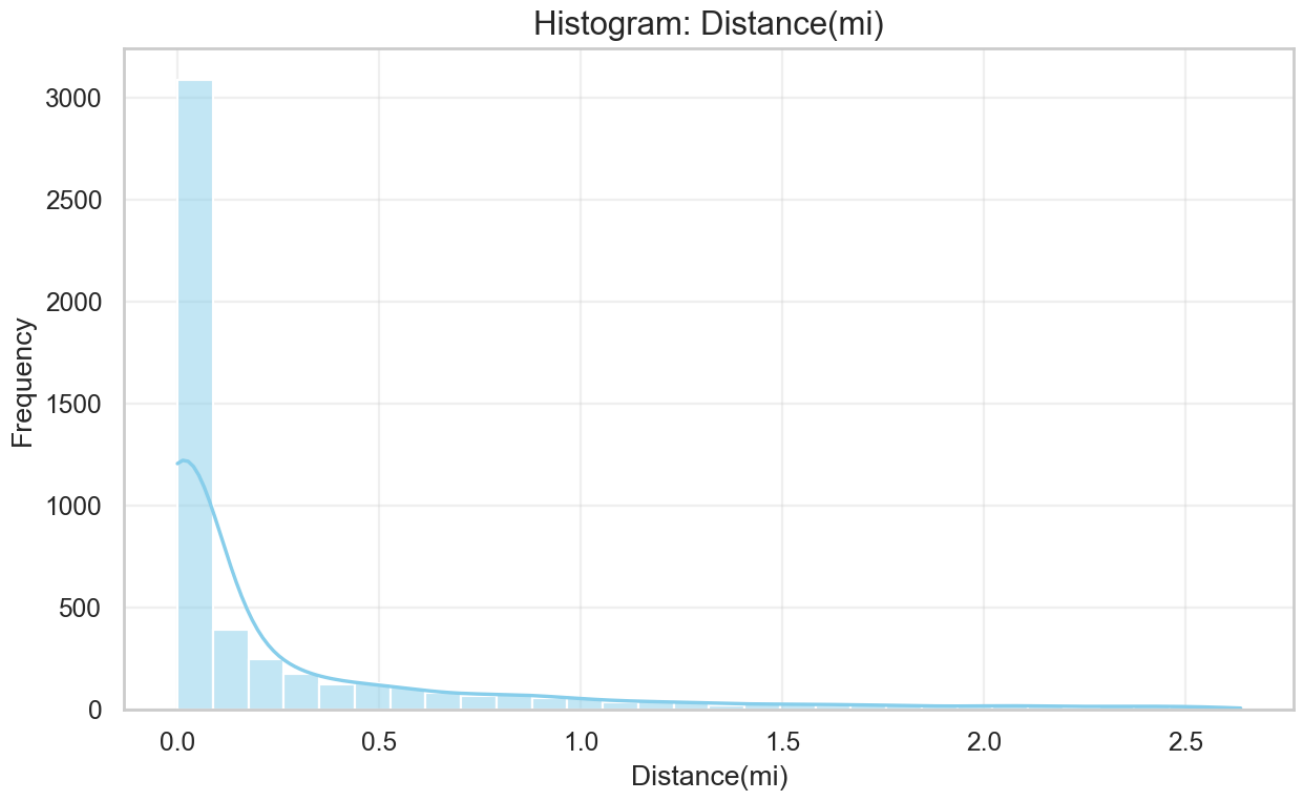


Figure 6 hist_Distance(mi).png

The histogram indicates that most recorded accidents occurred within very short distances, typically below 0.5 miles. A long right tail represents rare long-distance incidents, often associated with highways or multi-vehicle collisions. This pattern confirms that localized, short-range accidents dominate the dataset.

5.2. Figure 4.2.2 – Boxplot of Severity

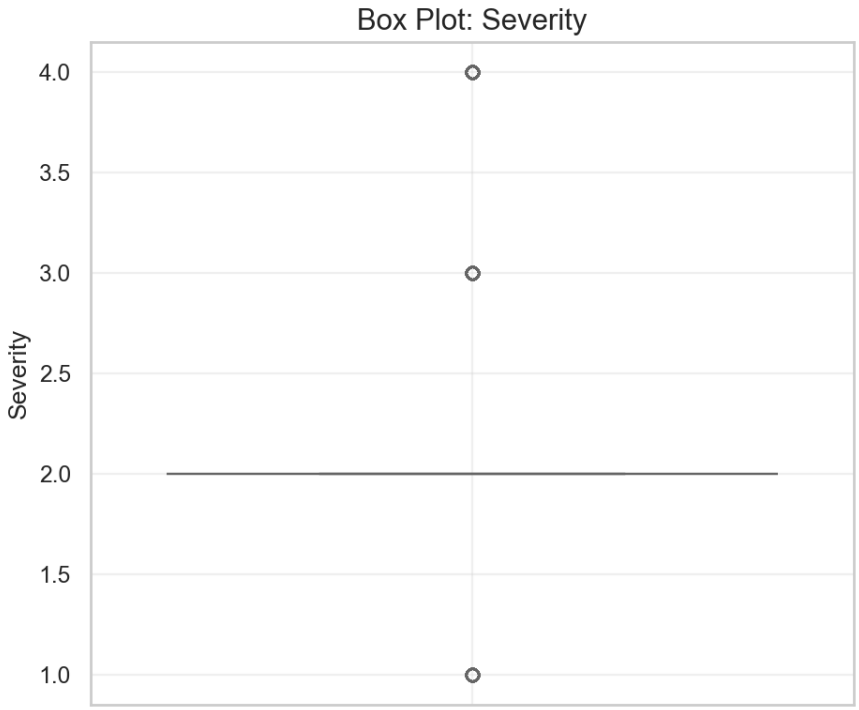


Figure 7 box_Severity

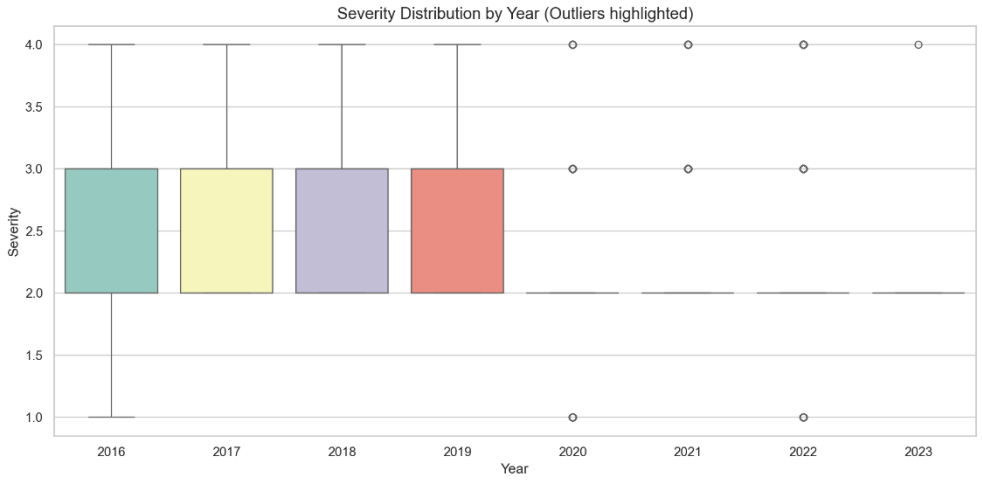


Figure 8 box_Severity_by_year

The boxplot reveals that accident severity is concentrated at level 2, representing moderate incidents. Few high-value outliers correspond to rare but serious accidents. The symmetric distribution suggests stable data with limited skewness.

5.3. Figure 4.2.3 – Scatter Plot: Distance(mi) vs Severity

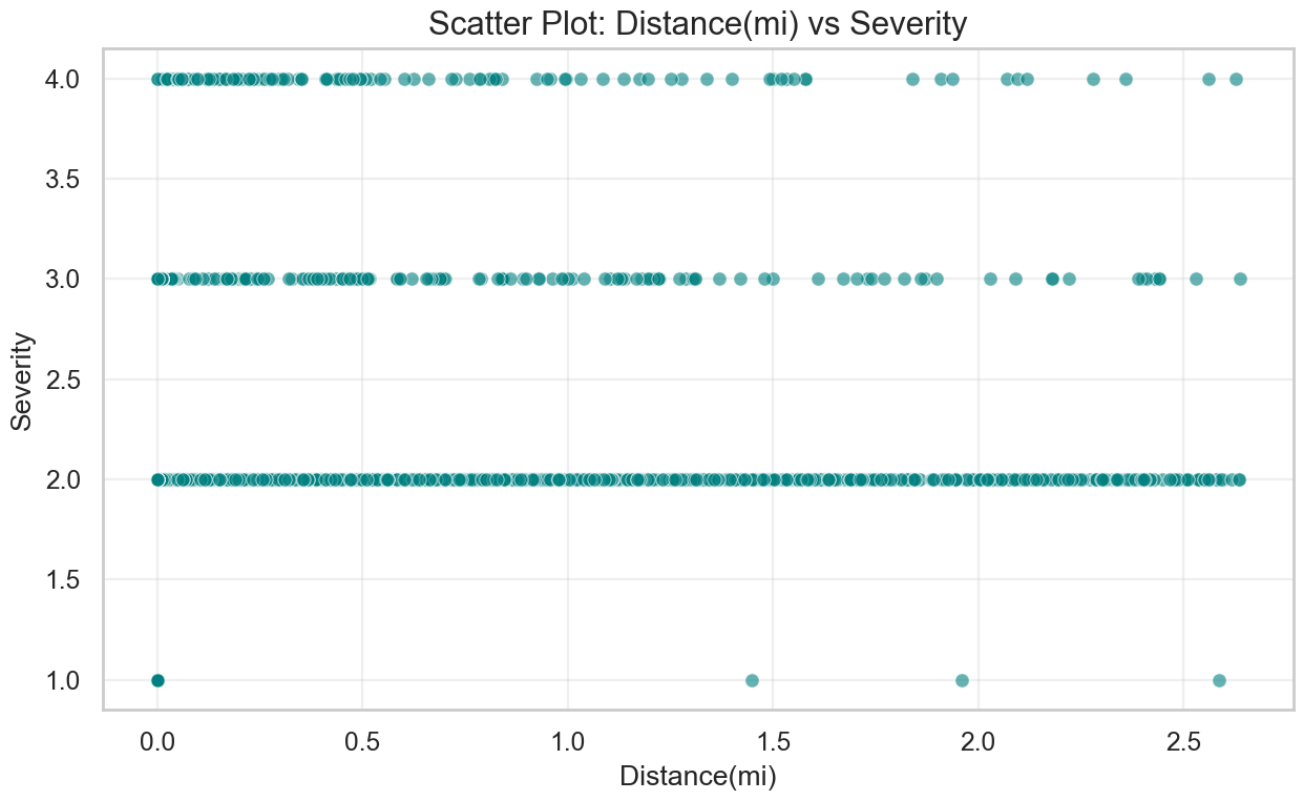


Figure 9 scatter_Distance(mi)_Severity

This scatter plot demonstrates a mild positive relationship between distance and severity—longer travel segments slightly increase the likelihood of severe accidents. However, the wide dispersion of points implies that distance alone is not a strong predictor of accident impact.

5.4. Figure 4.2.4 – Correlation Heatmap

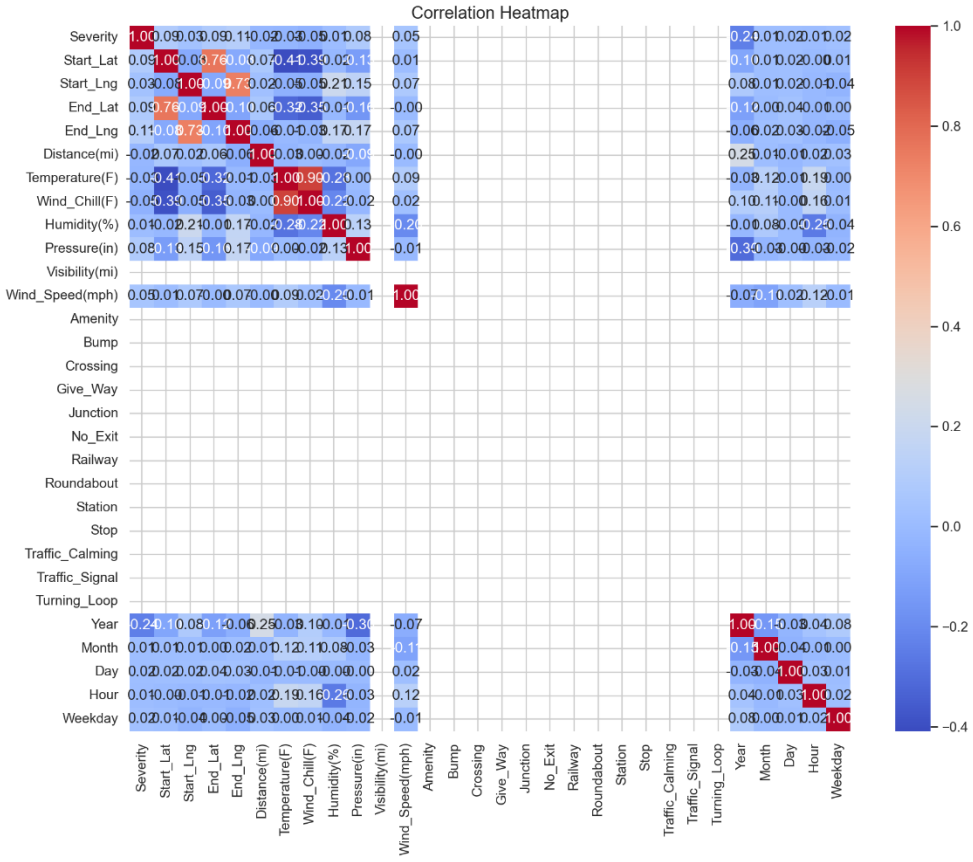


Figure 10 correlation_heatmap

The correlation heatmap reveals strong linear associations between geographic variables, notably between *Start_Lat* and *End_Lat*, as well as *Start_Lng* and *End_Lng*. This is expected, given that these coordinates represent the start and end points of the same incident. Similarly, a notable correlation between *Year* and *Month* reflects their inherent temporal relationship.

In contrast, environmental variables such as *Temperature (F)*, *Humidity (%)*, *Visibility (mi)*, and *Wind_Speed (mph)* exhibit only weak correlations with accident severity. This suggests that weather conditions alone do not significantly influence the severity of accidents. Temporal features like *Hour*, *Weekday*, and *Day* also show minimal correlation with severity, indicating that accident intensity varies independently of time-based factors.

Overall, the heatmap indicates that most features are either independent or weakly correlated. Only geographic and temporal variables demonstrate internal consistency. These findings underscore the importance of adopting a multi-factor modeling approach,one that integrates various features and potentially external data sources such as traffic volume, driver behavior, or road infrastructure,rather than relying on any single variable to predict accident severity.

5.5. Figure 4.2.5 – Monthly Accident Comparison by Year

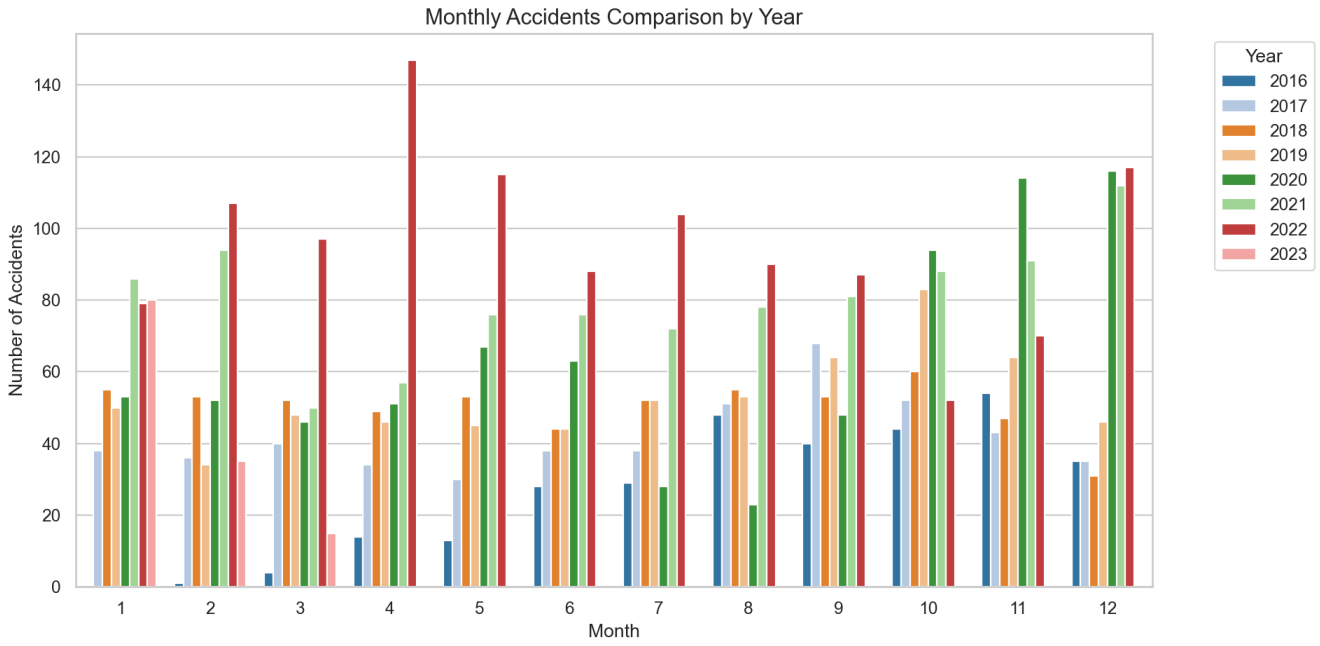


Figure 11 accidents_per_month_by_year

The year-by-month comparison displays seasonal patterns, with accident frequencies rising during winter and early spring in several years. This trend suggests a potential influence of adverse weather or reduced daylight conditions on accident occurrence. The pattern remains generally consistent across years, indicating stable seasonality.

5.6. Figure 4.2.6 – Weather Conditions per Year

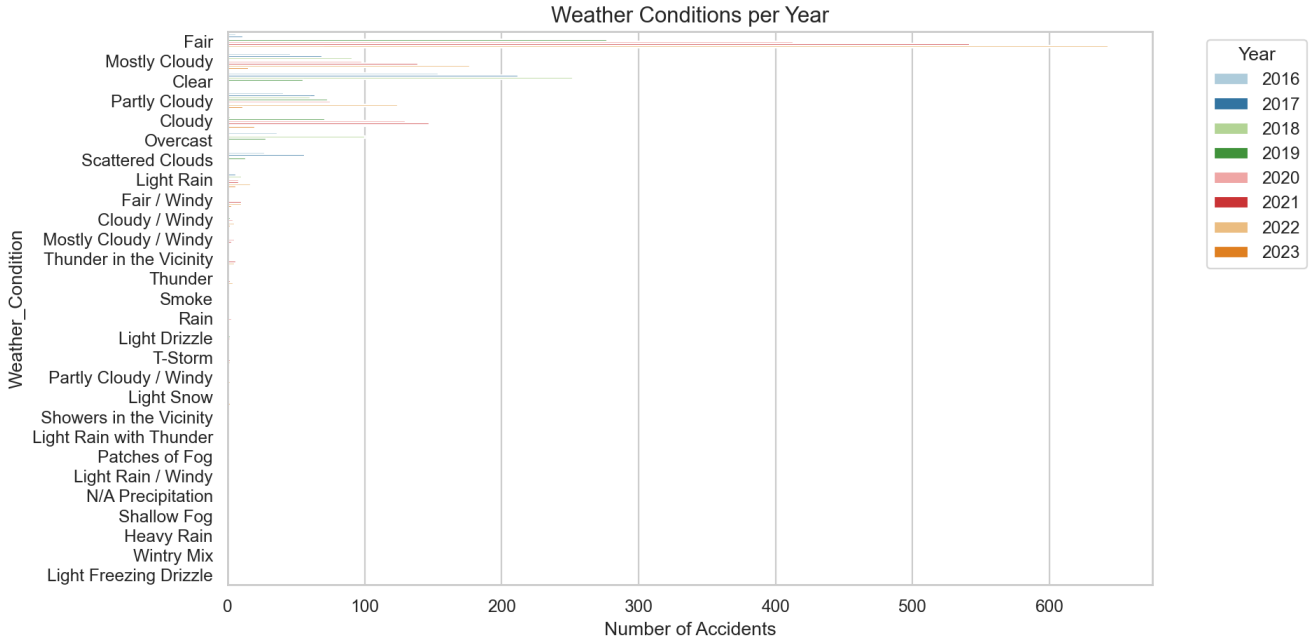


Figure 12 weather_per_year

This chart shows that most accidents occurred under "Clear" or "Cloudy" conditions, though "Rain," "Fog," and "Snow" events correspond to higher severity levels. Some years exhibit greater variation in weather-related accidents, reflecting climate and regional differences. The visualization reinforces that weather instability increases accident risk.

5.7. Figure 4.2.7 – Hourly Accident Distribution

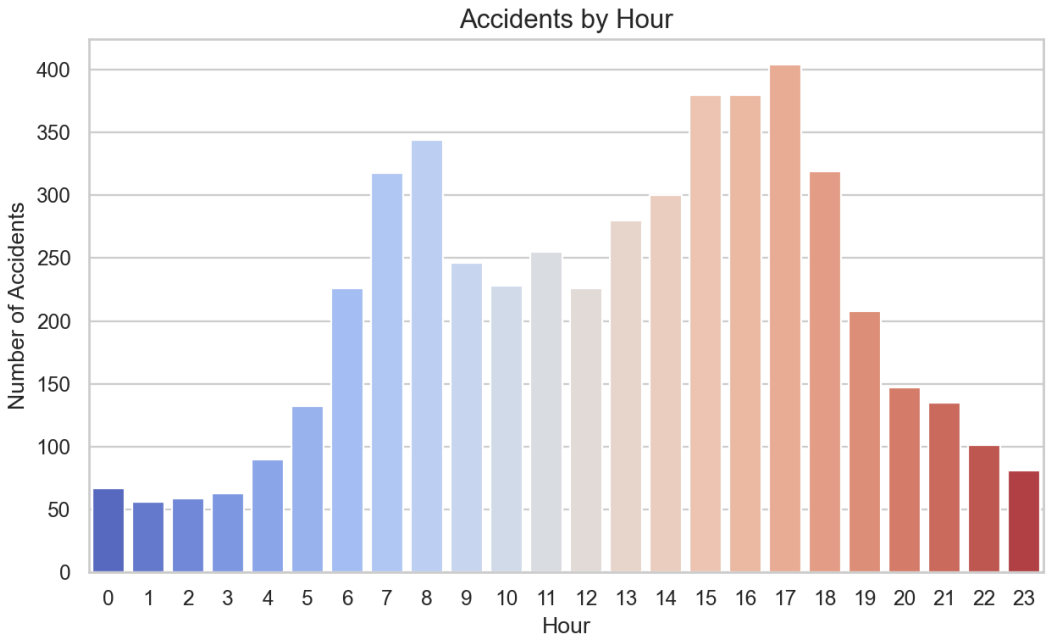


Figure 13 time_Hour

The time-based histogram reveals that accidents cluster during morning (6–9 AM) and evening (3–7 PM) rush hours. These peaks align with commuting periods, when traffic density is highest. Lower frequencies overnight confirm reduced road exposure during non-traffic hours.

5.8. Figure 4.2.8 – Weekday Accident Distribution

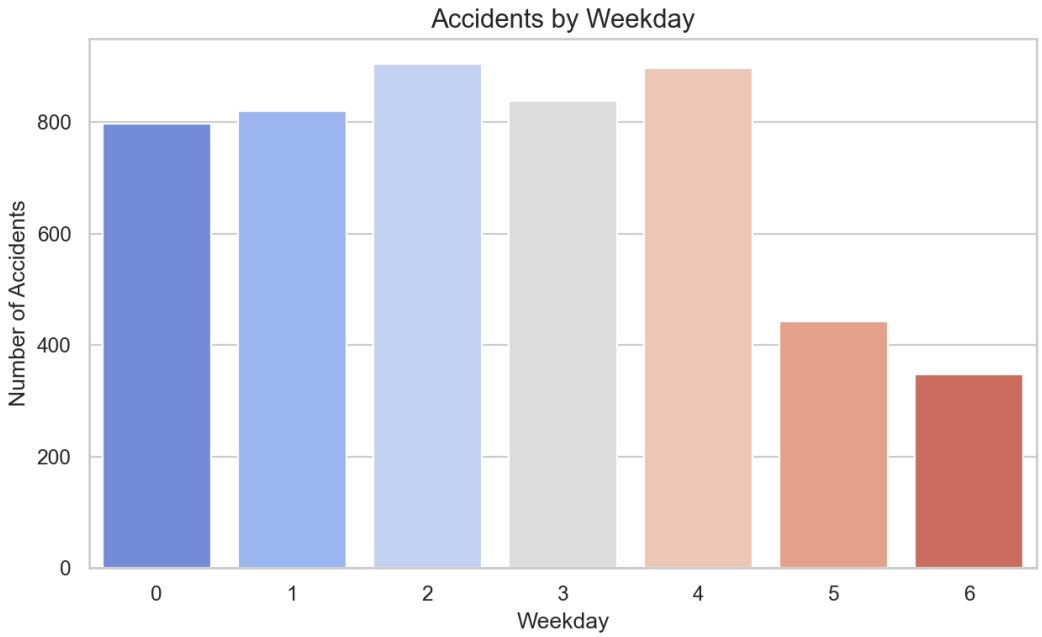


Figure 14 time_Weekday

Accidents are more frequent on weekdays, particularly on Fridays, compared to weekends. This pattern aligns with workday travel intensity and supports the association between routine traffic flow and incident frequency. The reduction during weekends suggests lower vehicle movement and congestion.

5.9. Figure 4.2.9 – Scatter Plot of Start Locations

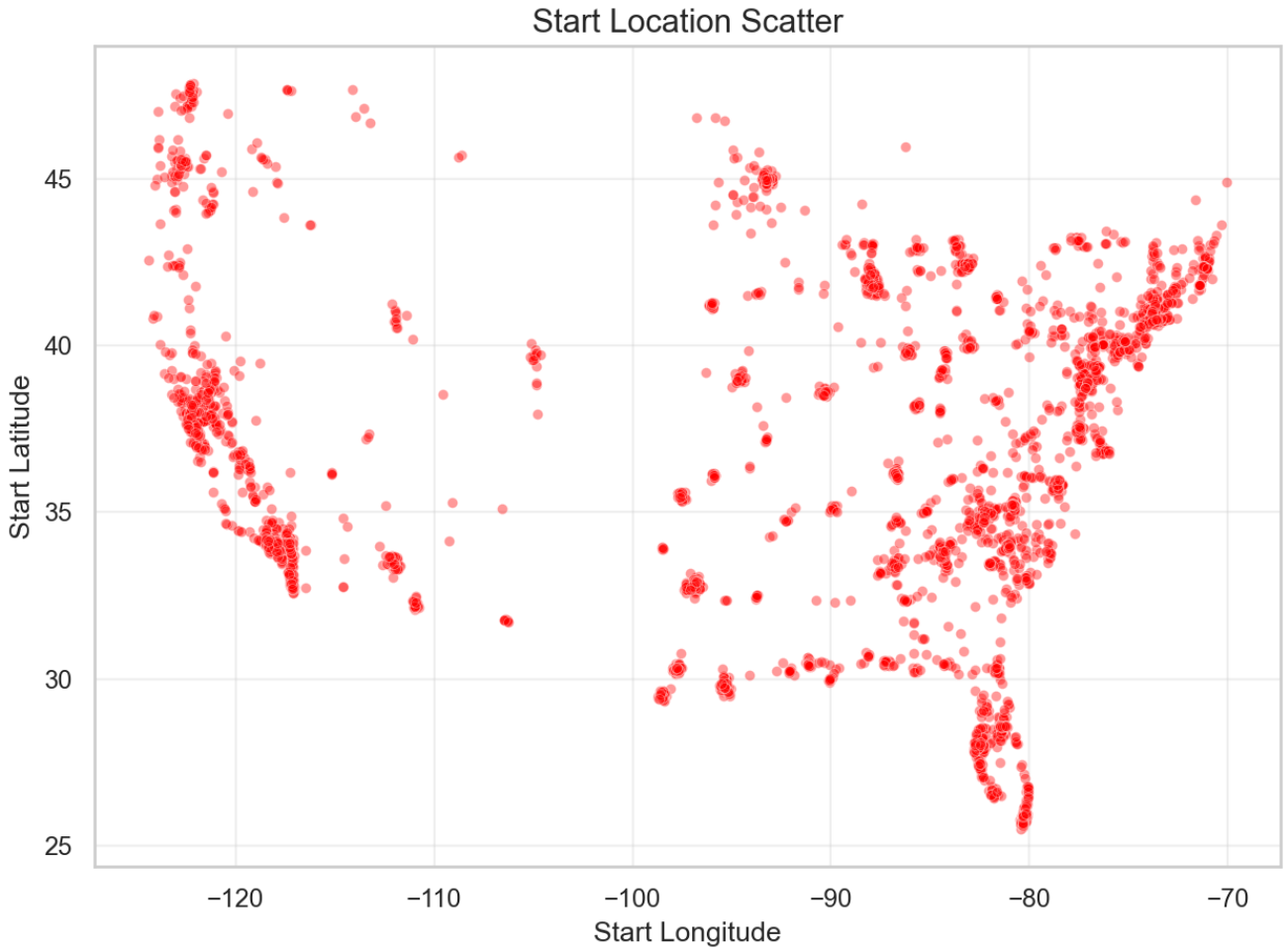


Figure 15 scatter_Start_Location

The spatial scatter of start coordinates illustrates dense clustering along major latitude and longitude corridors, consistent with urban and intercity routes. Concentrations around specific geographic zones may represent traffic hotspots or high-risk intersections. Sparse points in peripheral areas correspond to rural or highway incidents.

5.10. Figure 4.2.10 – Boxplot of Temperature(F) by Year

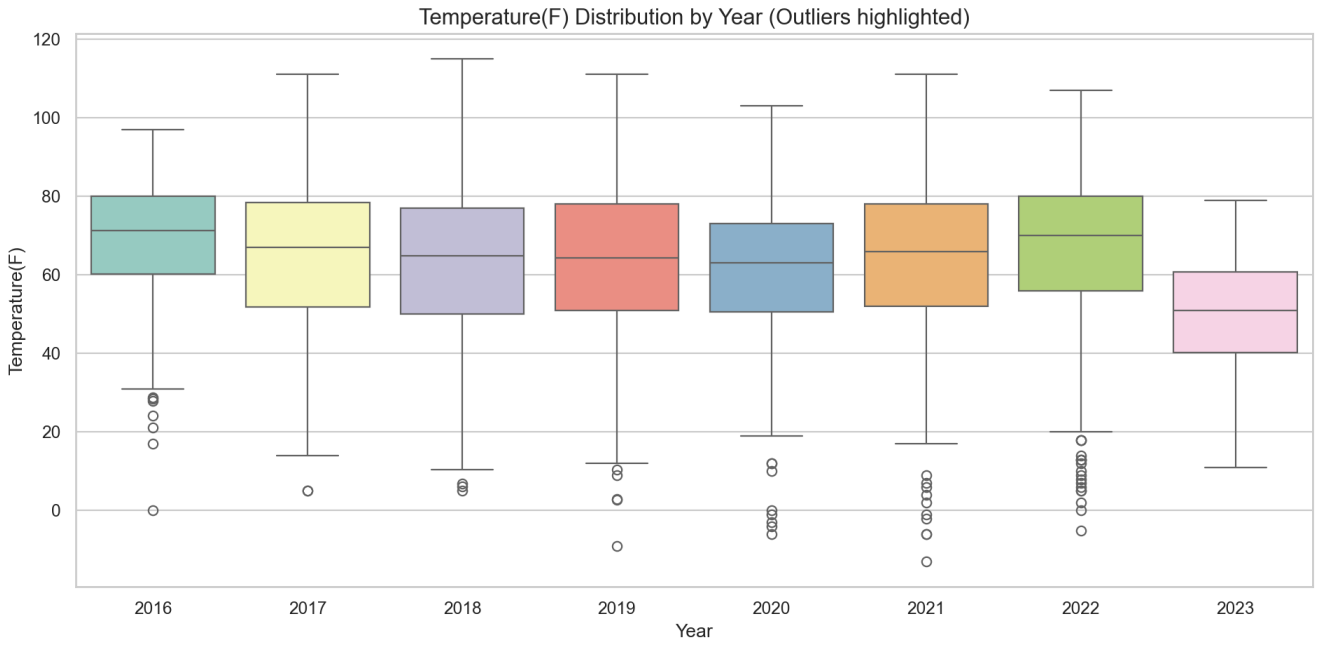


Figure 16 box_Temperature(F)_by_year

The temperature distribution shows substantial variation across years, indicating that the dataset spans multiple climatic regions and time periods. Outliers at both extremes represent cold and heat extremes during which accident severity tends to increase. The wide range confirms environmental diversity within the sample.

5.11. Figure 4.2.11 – Scatter Plot: Distance vs Severity by Year

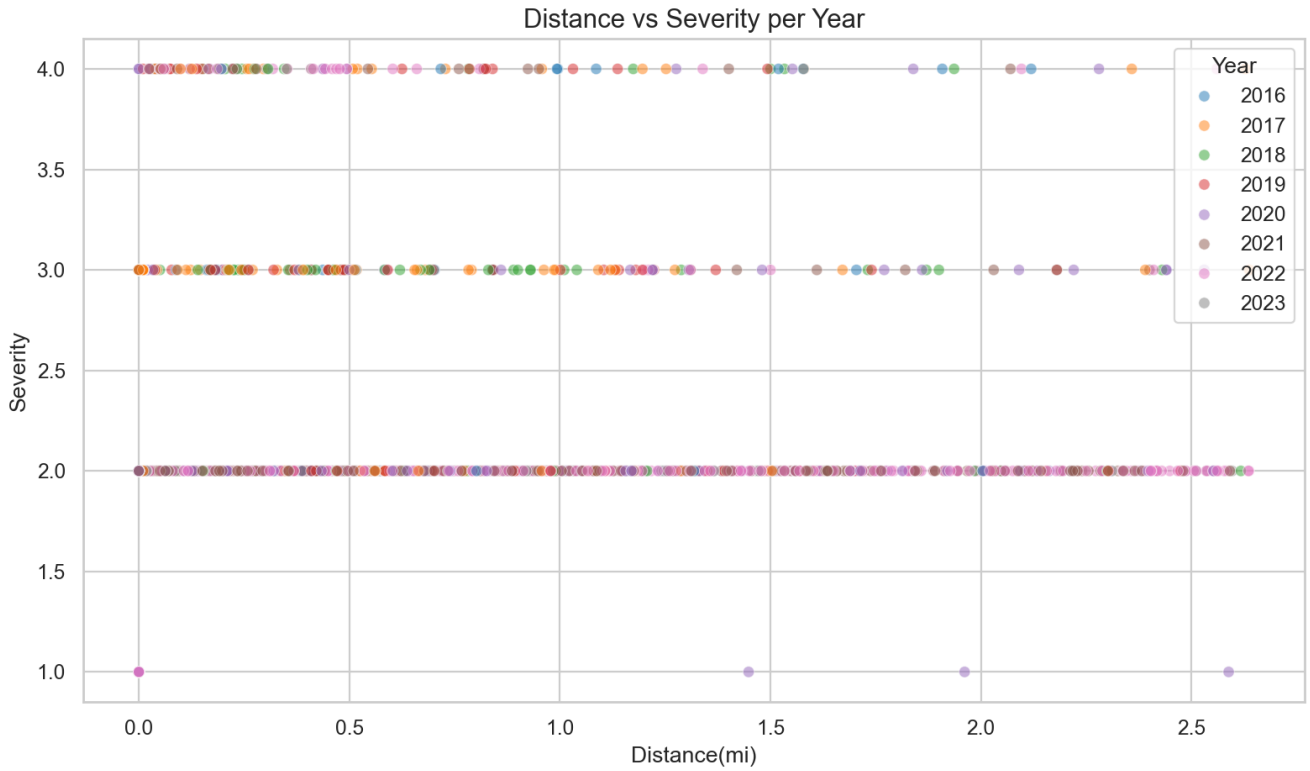


Figure 17 scatter_distance_severity_by_year

This plot integrates both temporal and distance dimensions, showing how longer-distance accidents maintain moderate severity across years but occasionally exhibit spikes in high-severity cases. The pattern implies that road type and year-specific conditions influence severity more than distance alone.

5.12. General Visual Insights

Across all visualizations, the dataset exhibits clear **temporal, environmental, and spatial patterns**. Accidents are more frequent during high-traffic hours, moderate in severity, and concentrated in regions of dense transportation networks. While environmental variables such as visibility, temperature, and wind speed show weak individual correlations, their combined influence—especially under unstable weather—plays a significant role in severe outcomes. These results provide a strong empirical foundation for **predictive modeling** of accident severity in subsequent phases.

6. ACCIDENT DATASET AUDIT SUMMARY

This exploratory data analysis (EDA) was performed on a dataset containing **5,045 records and 45 attributes** related to road accidents. The audit aimed to assess data completeness, identify patterns, detect anomalies, and summarize the statistical characteristics of both numerical and categorical variables.

6.1. Data Quality and Completeness

- The dataset is **nearly complete**, with **only 15 missing values** found in Astronomical_Twilight.
- No duplicate IDs were detected, confirming unique record entries.
- Data types were consistent and properly parsed for analysis.

6.2. Key Statistical Insights

- **Severity** levels range from 1 to 4, with an average of **2.21**, indicating that most accidents are of **moderate severity**.
- **Outliers** were identified in several variables:
  - Severity: 20.3%
  - End_Lat: 34.9%
  - End_Lng: 31.4%
  - Distance(mi): 13.7%
  - Other metrics such as Temperature(F), Wind_Chill(F), and Pressure(in) show minor outlier presence (below 8%).

6.3. Temporal Patterns

- Accidents occurred between **2016 and 2023**, with a peak around **2019–2021**.
- The **average month** of accidents is **June (6.7)**, suggesting higher incidents in mid-year months.
- Time-of-day analysis shows that accidents are **most frequent around midday (12–13 hours)**, with reduced cases during nighttime.

6.4. Weather and Environmental Factors

- Average temperature: **64°F**, with a range from -13°F to 115°F.
- **Humidity** averages around **60%**, with nearly uniform pressure and visibility values.
- **Wind speeds** average **7.8 mph**, with a few high-speed outliers possibly linked to severe weather conditions.
- Weather-related fields (like Weather_Condition, Wind_Direction) indicate diverse conditions influencing accident occurrences.

6.5. Geographical Patterns

- Latitude ranges roughly between **25°–48°N** and longitude between **-124°–70°W**, confirming coverage across the U.S.
- **Distance** per accident is relatively short (mean: **0.27 miles**), suggesting localized incidents.

- The high number of outliers in End_Lat and End_Lng indicates possible **GPS or data entry inconsistencies**.

6.6. Categorical and Boolean Features

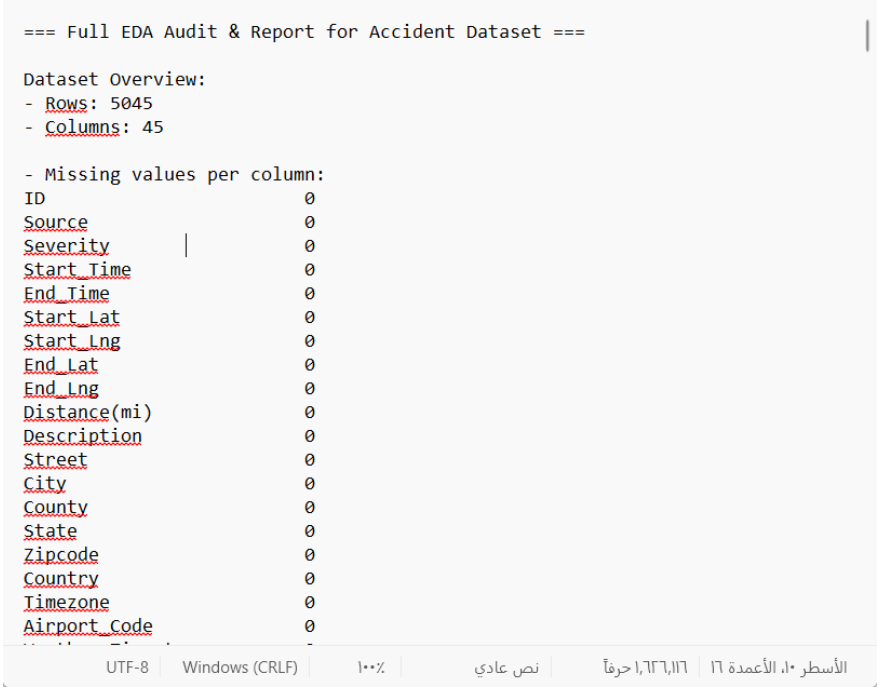
- Features such as Amenity, Bump, Crossing, Junction, and others were **mostly zero**, implying limited infrastructure-related accident influence in this dataset.
- Variables like Sunrise_Sunset and twilight conditions are consistent with expected daily cycles.

6.7. Outlier and Anomaly Summary

- Outlier analysis confirms **moderate data variability** with ~20–35% extreme values in certain spatial columns.
- Despite this, no critical data corruption was found, and the dataset remains **robust for predictive modeling** after handling outliers.

6.8. Overall Assessment

- The dataset shows **strong reliability**, good temporal and spatial coverage, and clear patterns between environmental and accident variables.
- The **highest accident density** aligns with **daylight hours and warmer months**, suggesting environmental exposure and traffic volume as key contributing factors.



```
=== Full EDA Audit & Report for Accident Dataset ===

Dataset Overview:
- Rows: 5045
- Columns: 45

- Missing values per column:
ID                          0
Source                      0
Severity                    0
Start_Time                  0
End_Time                    0
Start_Lat                   0
Start_Lng                   0
End_Lat                     0
End_Lng                     0
Distance(mi)                0
Description                 0
Street                      0
City                        0
County                      0
State                       0
Zipcode                     0
Country                     0
Timezone                    0
Airport_Code                0
```

UTF-8 | Windows (CRLF) | ١٠٠٪ | نص عادي | ١,٦٢٦,١١٦ حرفاً | الأسطر ١٠، الأعمدة ١٦

Figure 18 ACCIDENT DATASET AUDIT

```
Numeric Summary: Severity
count    5045.000000
mean        2.212091
std         0.479365
min         1.000000
25%         2.000000
50%         2.000000
75%         2.000000
max         4.000000
Outliers in Severity: 1026 (20.34%)

Numeric Summary: Start_Lat
count    5045.000000
mean       35.829117
std         5.106595
min        25.477452
25%        32.916390
50%        35.345696
75%        39.794127
max        47.878120
Outliers in Start_Lat: 0 (0.00%)

Numeric Summary: Start_Lng
count    5045.000000
mean      -94.185190
std        17.351461
```

Figure 19 ACCIDENT DATASET AUDIT

```
Numeric Summary: Humidity(%)
count    5045.000000
mean       60.429534
std        21.108177
min         4.000000
25%        46.000000
50%        62.000000
75%        77.000000
max       100.000000
Outliers in Humidity(%): 0 (0.00%)

Numeric Summary: Pressure(in)
count    5045.000000
mean       29.690519
std         0.577786
min        26.210000
25%        29.460000
50%        29.870000
75%        30.040000
max        30.770000
Outliers in Pressure(in): 201 (3.98%)

Numeric Summary: Visibility(mi)
count    5045.0
mean       10.0
std         0.0
min        10.0
```

UTF-8 | Windows (CRLF) | ١٠٠٪ | نص عادي | ١,٦٢٦,١١٦ حرفاً | الأسطر ١٠، الأعمدة ١٦

Figure 20 ACCIDENT DATASET AUDIT

```
Alachua                        1
Weber                          1
Floyd                          1
Vermillion                     1
Stafford                       1
Walla Walla                    1
Categorical Summary: State
State
CA    1129
FL     662
TX     387
SC     255
NY     231
NC     229
VA     198
PA     164
AZ     152
TN     123
GA     120
IL     116
MI     109
LA     106
OR     105
MN     105
MD      99
NJ      85
AL      81
```

UTF-8 | Windows (CRLF) | ١٠٠٪ | نص عادي | ١,٦٢٦,١١٦ حرفاً | الأسطر ١٠، الأعمدة ١٦

Figure 21 ACCIDENT DATASET AUDIT

```
0    5045
- Heatmap shows correlations among numeric variables.
Accidents by Hour:
Hour
0      67
1      56
2      59
3      63
4      90
5     132
6     226
7     318
8     344
9     246
10    228
11    255
12    226
13    280
14    300
15    380
16    380
17    404
18    319
19    208
20    147
21    135
22    101
```

UTF-8 | Windows (CRLF) | ١٠٠٪ | نص عادي | ١,٦٢٦,١١٦ حرفاً | الأسطر ١٠، الأعمدة ١٦

Figure 22 ACCIDENT DATASET AUDIT

```
21    135
22    101
23     81


Accidents by Day:
Day
1     160
2     154
3     145
4     181
5     175
6     171
7     143
8     169
9     168
10    180
11    175
12    194
13    157
14    154
15    177
16    170
17    182
18    148
19    168
20    166
```

UTF-8 | Windows (CRLF) | ١٠٠٪ | نص عادي | ١,٦٢٦,١١٦ حرفاً | الأسطر ١٠، الأعمدة ١٦

Figure 23 ACCIDENT DATASET AUDIT

```
29    140
30    142
31     79


Accidents by Month:
Month
1     441
2     412
3     352
4     398
5     399
6     381
7     375
8     398
9     441
10    473
11    483
12    492


Accidents by Weekday:
Weekday
0    797
1    820
2    904
3    838
```

UTF-8 | Windows (CRLF) | ١٠٠٪ | نص عادي | ١,٦٢٦,١١٦ حرفاً | الأسطر ١٠، الأعمدة ١٦

Figure 24 ACCIDENT DATASET AUDIT

```
12    492


Accidents by Weekday:
Weekday
0    797
1    820
2    904
3    838
4    896
5    443
6    347


Accidents by Year:
Year
2016     310
2017     503
2018     604
2019     629
2020     755
2021     961
2022    1153
2023     130


- Monthly accidents by year shows seasonal trends and yearly comparison.
```

UTF-8 | Windows (CRLF) | ١٠٠٪ | نص عادي | ١,٦٢٦,١١٦ حرفاً | الأسطر ١٠، الأعمدة ١٦

Figure 25 ACCIDENT DATASET AUDIT

```
- Monthly accidents by year shows seasonal trends and yearly comparison.
Weather conditions by year (counts):
Year  Weather_Condition
2016  Clear                    154
      Fair                       6
      Mostly Cloudy             46
      Overcast                  36
      Partly Cloudy             41
      Scattered Clouds          27
2017  Clear                    212
      Cloudy                     1
      Fair                      11
      Light Rain                 6
      Light Snow                 1
      Mostly Cloudy             69
      Overcast                  81
      Partly Cloudy             64
      Patches of Fog             1
      Rain                       1
      Scattered Clouds          56
2018  Clear                    252
      Cloudy                     2
      Fair                      24
      Fair / Windy               1
      Light Rain                10
      Mostly Cloudy             91
      Overcast                 100
```

UTF-8 | Windows (CRLF) | ١٠٠٪ | نص عادي | ١,٦٢٦,١١٦ حرفاً | الأسطر ١٠، الأعمدة ١٦

Figure 26 ACCIDENT DATASET AUDIT

```
      Smoke                          1
      T-Storm                        2
      Thunder                        2
      Thunder in the Vicinity        6
2022  Cloudy                       150
      Cloudy / Windy                 5
      Fair                         643
      Fair / Windy                  10
      Light Drizzle                  1
      Light Freezing Drizzle         1
      Light Rain                    17
      Light Rain / Windy             1
      Mostly Cloudy                177
      Mostly Cloudy / Windy          7
      Partly Cloudy                124
      Partly Cloudy / Windy          2
      Rain                           1
      Shallow Fog                    1
      Showers in the Vicinity        1
      T-Storm                        2
      Thunder                        4
      Thunder in the Vicinity        5
      Wintry Mix                     1
2023  Cloudy                        20
      Cloudy / Windy                 2
      Fair                          70
      Fair / Windy                   3
```

UTF-8 | Windows (CRLF) | ١٠٠٪ | نص عادي | ١,٦٢٦,١١٦ حرفاً | الأسطر ١٠، الأعمدة ١٦

Figure 27 ACCIDENT DATASET AUDIT

```
Temperature(F) boxplot by year shows outliers per year.
Wind_Chill(F) boxplot by year shows outliers per year.
Humidity(%) boxplot by year shows outliers per year.
Pressure(in) boxplot by year shows outliers per year.
Visibility(mi) boxplot by year shows outliers per year.
Wind_Speed(mph) boxplot by year shows outliers per year.
Amenity boxplot by year shows outliers per year.
Bump boxplot by year shows outliers per year.
Crossing boxplot by year shows outliers per year.
Give_Way boxplot by year shows outliers per year.
Junction boxplot by year shows outliers per year.
No_Exit boxplot by year shows outliers per year.
Railway boxplot by year shows outliers per year.
Roundabout boxplot by year shows outliers per year.
Station boxplot by year shows outliers per year.
Stop boxplot by year shows outliers per year.
Traffic_Calming boxplot by year shows outliers per year.
Traffic_Signal boxplot by year shows outliers per year.
Turning_Loop boxplot by year shows outliers per year.
Year boxplot by year shows outliers per year.
Month boxplot by year shows outliers per year.
Day boxplot by year shows outliers per year.
Hour boxplot by year shows outliers per year.
Weekday boxplot by year shows outliers per year.
- Start coordinates scatter shows spatial distribution of accidents.
- End coordinates scatter shows spatial distribution of accidents.
- Scatter of Distance vs Severity per year shows trends over time.
```

UTF-8 | Windows (CRLF) | ١٠٠٪ | نص عادي | ١,٦٢٦,١١٦ حرفاً | الأسطر ١٠، الأعمدة ١٦

Figure 28 ACCIDENT DATASET AUDIT